Initial Project Planning Template

Date	1 July 2026
Team ID	SWTID-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Epic 1: Environment Setup and Dependency Configuration	US-1	As a developer, I can create a Python virtual environment and install all required dependencies (Streamlit, Whisper, Sentence-Transformers, Librosa, ReportLab) so the project runs without errors.	1	High	Palak Suthar	30 July 2026	1 July 2026
Sprint-1	Epic 1: Environment Setup and Dependency Configuration	US-2	As a developer, I can organize the project into modular files (app.py, speech_to_text.py, audio_utils.py, semantic_eval.py, scoring_engine.py, report_generator.py) so,the codebase is maintainable.	1	Medium	Palak Suthar	30 July 2026	1 July 2026
Sprint-1	Epic 1: Environment Setup and Dependency Configuration	US-3	As a developer, I can launch the Streamlit application locally and confirm the UI loads correctly so development can proceed.	1	High	Harshita Soni	30 July 2026	1 July 2026
Sprint-2	Epic 2: Core Logic Development (Speech Understanding & Evaluation Engine)	US-4	As a user, I can upload an audio file and have it transcribed to text using OpenAI Whisper so my spoken explanation is captured.	2	High	Palak Suthar	1 July 2026	2 July 2026

Sprint-2	Epic 2: Core Logic Development (Speech Understanding & Evaluation Engine)	US-5	As a user, I can have my transcribed explanation compared against a reference concept using Sentence-BERT so I get a semantic similarity score.	1	High	Neelam Kumari	1 July 2026	2 July 2026
Sprint-2	Epic 2: Core Logic Development (Speech Understanding & Evaluation Engine)	US-6	As a user, I can have my audio analyzed for filler words, pauses, and energy so I receive a fluency and final understanding score.	2	High	Palak Suthar	1 July 2026	2 July 2026
Sprint-3	Epic 3: Streamlit UI Implementation and User Interaction	US-7	As a user, I can see a clean interface showing the reference concept, an audio upload control, and an Analyze button so I can start an evaluation.	3	High	Swati Yadav	1 July 2026	2 July 2026
Sprint-3	Epic 3: Streamlit UI Implementation and User Interaction	US-8	As a user, my transcript, scores, and audio features persist on screen via session state while I review results, without needing a database.	3	Medium	Neelam Kumari	1 July 2026	2 July 2026
Sprint-3	Epic 3: Streamlit UI Implementation and User Interaction	US-9	As a user, I can view my waveform, evaluation summary table, and download a PDF report so I have a record of my results.	3	High	Palak Suthar	1 July 2026	2 July 2026
Sprint-4	Epic 4: Testing, Optimization, and Deployment	US-10	As a developer, I can test the full pipeline (upload -> transcription -> scoring -> PDF) across multiple audio samples so I can confirm correctness.	2	High	Palak Suthar	2 July 2026	3 July 2026
Sprint-4	Epic 4: Testing, Optimization, and Deployment	US-11	As a developer, I can cache model loading and measure processing time so the app performs reliably on repeated use.	2	Medium	Palak Suthar	2 July 2026	3 July 2026
Sprint-4	Epic 4: Testing, Optimization, and Deployment	US-12	As a developer, I can prepare the app for local/cloud deployment and perform a final end-to-end review before submission.	2	Medium	Palak Suthar	2 July 2026	3 July 2026